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Renewable energy, nonrenewable energy, and environmental quality nexus: An investigation of the N-shaped Environmental Kuznet Curve based on six environmental indicators

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Abstract

The current research evaluated the role of non-renewable and renewable energy consumption in environmental sustainability following the N-shaped Environmental Kuznets Curve framework controlling financial development, population density, and composite trade share. Unlike previous studies, this research used six indices to depict environmental deterioration and environmental quality. More precisely, pressures on nature, ecological footprints, environmental vulnerability, and adjusted net saving are employed to represent environmental deterioration, while environmental performances and environmental sustainability are adopted to proxy environmental quality. The Dynamic Seemingly Unrelated Regression Equations were used on panel data from 1994 to 2019 in the selected Organization of the Petroleum Exporting Countries. The findings revealed the N-shaped linkage between per capita income and environmental deterioration indicators, such as ecological footprints, adjusted net saving, pressures on nature, and environmental vulnerability. Regarding the linkage between environmental quality indicators, including environmental performances and environmental sustainability, and per capita income, an inverted N-shaped Environmental Kuznets Curve is verified. Considering all six environmental indicators, non-renewable energy tends to deteriorate the environment, while the use of renewable energy promotes environmental quality. Moreover, population density, financial development, and composite trade share raise environmental deterioration. The results are vital for achieving a green economy with better environmental quality.

Keywords: Environmental indicators, Economic growth, Energy consumption, N-shaped EKC

1. Introduction

In recent decades, the energy and economic development association remains a hot topic in the environmental economics literature because energy and natural resources contribute significantly to economic development [1]. In addition, considerable resources are needed to guarantee economic growth, which usually decreases environmental quality (EQ) [2]. There is a strong association between different levels of economic activities (economic development) and energy usage. Energy is a necessary input for all economic functions, such as transportation of goods, electricity generation, trade, agriculture, and basic social services (education and health). Thus, energy contributes significantly to establishing the foundation for the advancement and development of every economy [3, 4, 5]. Energy has been assumed as the determining factor for economic development along with other production factors and its significance has continuously increased. The energy sector has more interaction with different economic sectors due to the growing demand for energy on account of rapid economic development and industrialization [6, 7].

Energy has the largest contribution to changing environmental conditions, compared to other factors [8, 6]. Regarding adverse environmental consequences of energy, Myers and Kent [9] believe that most of the greenhouse gas emissions, which are in the form of carbon dioxide (CO₂) are caused by fossil fuels combustion. Accordingly, there is a close relationship between the policies of the energy sector and environmental sustainability. The adverse impacts of energy on our environment require that countries that rely on fossil energy switch toward the use of renewable energy because renewable energy is believed to be environmentally friendly [10, 11]. It is also worth mentioning that renewable energy resources lessen CO₂ emissions and gives us help in protecting the environment [12, 5]. Therefore, the renewable energy resources seem keys to resolve energy securities problems and environmental pollution [13]. In this context, this research probes the environmental consequences of nonrenewable and

renewable energy usage in the Organization of the Petroleum Exporting Countries (OPEC) economies.

Since this issue has become one of the most important concerns for human societies, many empirical studies have evaluated the effect of economic development and energy consumption on the quality of the environment in the context of the Environment Kuznets Curve (EKC) hypothesis, which explains the inverted U-shaped association of economic development and the degradation of the environment [14, 6]. Given that a large part of energy demand is met by fossil fuels and economic growth needs energy consumption which produces greenhouse gas emissions and air pollution, economic growth seems to increase environmental pollution at the first glance. Nevertheless, the EKC hypothesis explains that economic development causes a surge in pollution only during the early stages of growth. Meanwhile, environmental issues become more important and the increasing trend of energy consumption and environmental pollution are reduced due to energy efficiency, environmental regulations, and an increase in public awareness at later stages of development [15, 1].

More precisely, environmental degradation increases at the first stage of economic development due to the scale effect. Nonetheless, the economic structure of countries changes with more development, and the service sector causes less emission in comparison with the industrial sector due to the composition effect. After this, technological enhancement (technique effect) may assist countries in alleviating environmental degradation. As a result, environmental pollution will finally decline via these two effects [16]. Although, the technological obsolescence impact transforms the inverted U-shaped into an N-shaped relationship because this is attributable to obsolescence and depreciation in technology and capital that result in increased pollution [17]. For this reason, instead of traditional EKC, the N-shaped EKC is becoming popular in recent literature. In this context, this research will empirically unveil a possible N-shaped relationship through technological obsolescence effect between economic growth and environmental degradation.

However, outcomes of the previous investigation on this topic are not homogenous and differences have been detected in the relationships between growth and environmental quality in the context of the EKC hypothesis (for more details, see Figure 3 in the “Review of previous empirical literature” section). This heterogeneity can be the consequence of different environmental indices employed in the studies because environmental indicators can play an important and decisive role in shaping the relationships between the environment and economic development [18]. According to these empirical studies, various economic development levels in different regions and countries may have different and contrasting effects on the environment. Hence, there is a need to perform an investigation that could employ different environmental indicators to analyze the effect of renewable and nonrenewable in terms of the N-shaped EKC.

This study analyzes the role of nonrenewable and renewable energy in terms of N-shaped EKC using six environmental indicators in selected OPEC countries due to the increasing trend of economic growth and the decreasing quality of the environment in these countries. It is observed that achieving desirable economic growth in the selected OPEC economies has been led to adverse consequences for the environment as shown in Figures 1 and 2.

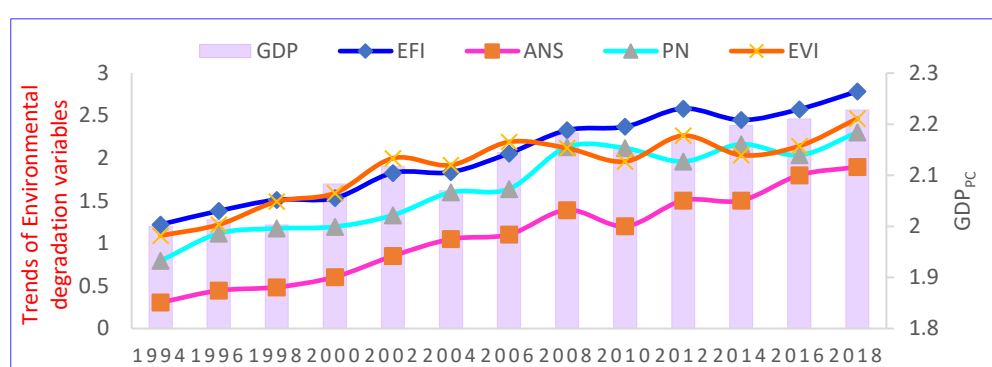


Figure 1 Trend of environmental deterioration related to EFI, ANS, PNI and EVI from the OPEC economies’ gross domestic product per capita (GDP_{pc})

Source: World Bank [19]

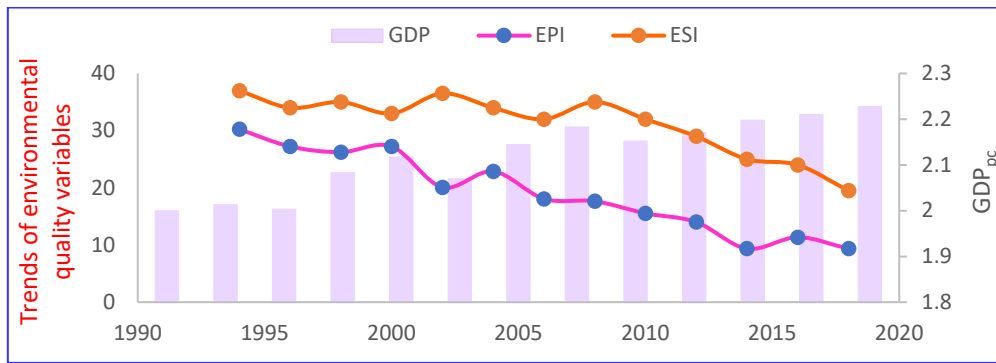


Figure 2 Trend of environmental quality related to EPI and ESI
from the OPEC economies' gross domestic product per capita (GDP_{pc})

Source: World Bank [19]

These trends show that not only OPEC members have turned to the wasteful use of natural resources to achieve higher economic growth as they have abundant natural resources (e.g., oil resources, fossil resources, and mineral resources), but also, they have welcomed pollutant industries aimed at expanding the industrial sectors [14].

In the light of the foregoing concerns and concerning the important role of energy and economic development in environmental quality and the challenging nature of this subject in empirical literature due to some important reasons (e. g., differences in data, evaluated countries, research methodology, and type of environmental indices used as dependent variables), it is the key issue for OPEC to dig deeper into the role of non-renewable energy consumption (NEC), renewable energy consumption (REC), and per-capita GDP on the quality of the environment considering the N-shaped EKC controlling financial development (FD), population density (POPD) and composite trade share (CTS).

To this end, the panel data from 1994 to 2019 are selected to explore the abovementioned variables connection using Dynamic Seemingly Unrelated Regression Equations (DSURE) technique, not only in terms of inversed N-shaped EKC but also regarding to the use of six environmental indices simultaneously, including pressures on nature, ecological footprints, environmental vulnerability, and adjusted net saving as proxies of environmental deterioration as well as environmental performances and environmental sustainability as proxies of

environmental quality which have not been conducted yet in the previous empirical research scholars in OPEC countries. The simultaneously employing these indicators can disclose a departure from the traditional EKC theory. Consequently, this study is a crucial step in closing literature gap.

Accordingly, in this study, we attempt to focus upon the following principal research questions: (i) To scrutinize whether the N-shaped EKC will be verified for all environmental indices. (ii) To test whether NEC and REC have any linkage with the environmental indices. (iii) To test the linkages of FD and CTS with the environmental indices. We believe that understanding these questions are of great value for providing pivotal insights for policymakers in the selected OPEC economies.

The rest of the paper is sectioned as follows: The review of the previous empirical literature is presented in section 2. In section 3, research data and methodology are explained, while sections 4 discusses empirical results followed by the conclusion and policy recommendations (section 5). Finally, limitations and future directions are presented in section 6.

2. Review of previous empirical literature

The EKC hypothesis proposed by the Grossman and Krueger [20, 21] became a vital framework to study the linkage between the progress of economies and environmental indicators in the presence of important driving forces of environmental degradation. This hypothesis explains that economic expansion stimulates environmental pollution due to the scale effect which demands more production generally at the cost of the environment [22]. However, economies that are dependent on the agriculture sector transform into more industrial-based economies over time. Indeed, industrialization intensifies environmental pollution; however, structural transformation expands the service sector, and with time, the service sector becomes the leading sector of the economy. Evidently, the service sector emits less pollution compared to the other two sectors [23]. Besides, higher development levels

promote environmental awareness, research and development, green technologies, clean energies, and stringent environmental regulations, which lessen environmental pollution [24]. Generally, the technique effect dominates at higher stages of development. Consequently, early stages of growth intensify environmental pollution but higher levels of economic growth decrease pollution levels resulting in an inverted U-shaped connection between growth and environmental pollution, which is famous as the EKC.

EKC is a widely adopted framework in environmental economics literature; however, recently, some studies argued that after the technique effect, the technological obsolescence effect emerges which converts the U-Shaped linkages into an N-shaped association since depreciation and obsolescence in technology mounts environmental pollution [17]. Thus, the scale effect can re-emerge and overpower the impacts of composition and technique effects leading to one more turning point, which forms an N-shaped curve. This study intends to explore the N-shaped EKC by considering six environmental indicators. It is vital to briefly explain the environmental indicators before discussing the past literature.

In the literature, ecological footprint (EFI) is used as a comprehensive measure of environmental degradation. EFI estimates the pressure of resource consumption in terms of cropland, fishing grounds, grazing land, forestland, carbon footprint, and built-up land footprint [25]. The adjusted net savings (ANS) indicator is composed of net national savings inclusive of education expenditure after reducing mineral depletion, energy depletion, net forest depletion, and CO₂ and other emissions damages. The pressures on nature (PNI) indicator considers the damages caused by CO₂, depletion of minerals, energies, and net forest [26]. The environmental performance (EPI) indicator measures the environmental health as well as the sustainability of nations by focusing on factors related to environmental health and ecosystem vitality [27]. The environmental sustainability (ESI) indicator ranks the nations in terms of their ability to protect the environment in future years. The environmental vulnerability (EVI) indicator depicts the vulnerability of the environment in various countries. It provides valuable

information about the processes that can adversely impact the sustainable development of nations [28].

Now, we proceed to the review of past literature. In this section, first of all, we briefly reviewed six environmental indices used by previous studies as environmental indicators in economic models. Afterwards, experimental results associated with the effects of economic variables (per capita GDP, REC, NEC, FD, POPD and TO) on EQ are briefly presented. Multiple indices were described for EQ and used in environmental-economic models. Finally, the literature gaps are presented.

Given the necessity of developing environmental indices for policy-makers in line with sustainable principles, different stages are considered for measuring and evaluating environmental outcomes of economic growth through the expansion of environmental criteria and indices [29], and various indices are applied in studies. According to the various environmental indices employed in these researches, the discrepancy in the results can be attributed to these studies' types of environmental indicators since the type of environmental indicators as a variable addressing different aspects of environmental pollution can play a critical and decisive role in the kind of relationship.

In many studies, only one indicator is used to show the state of the environment. For instance, EFI is used in many studies [29, 8, 30, 2], EPI is applied in some empirical works [18, 31, 32], ANS is employed in the empirical models of some academic works [33, 34, 35]; ESI is adopted in the empirical models of some experimental works [36, 37], EVI is employed in some investigations [38, 28], and PN is also used by the study of Chen et al. [39] and Asici [26]. Furthermore, we can refer to SO₂ in study of Bakhsh et al. [12] and CO₂ in studies of Ali et al. [40] and Bakhsh et al. [13].

The disputable point that should be considered in these studies is that in all these studies, only one environmental index and criterion was used. Since these indicators alone show a part of the overall damage to the environment and each of them cannot be considered as a general

and comprehensive environmental indicator, the failure to use these indicators as a dependent variable simultaneously can be considered as one of the existing gaps in past empirical studies. In other words, the literature's lack of consensus might be attributed to this point of the differences in indicators.

Different studies have assessed the correlation between the aforementioned economic variables and the quality of environment. Overall, it is possible to divide these studies into four groups, the first examined the interaction between GDP and EQ within the framework of the EKC hypothesis. Table 1 presents a list of recent studies on the EKC hypothesis.

Table 1. Studies on the EKC hypothesis with varying results

Author/s	Period	Result
Shahzad et al. [1]	1971-2016	EKC
Sharif et al. [3]	1965-2017	EKC
Asici [26]	1970-2008	EKC
Ozcan et al. [41]	2000-2013	EKC
Dogan et al. [42]	1980-2014	U-shaped
Shah et al. [43]	1980-2017	U-shaped
Yilanci and Pata [44]	1965-2016	U-shaped
Destek et al. [45]	1980-2013	U-shaped
Danish and wang [35]	1992-2013	N-shaped
Özokcu and Özdemir [46]	1980-2010	N-shaped
Anwar et al. [47]	1991-2018	Inverted N-shaped
Allard et al. [48]	1994-2012	Inverted N-shaped
Baek [49]	1960–2010	Positive
Nasreen et al. [50]	1980–2012	Negative

The first group of studies evaluated the association of economic development and EQ via the EKC framework. The first authors [20] who used the EKC framework, which was further used by other authors, including Panayotou [16], Dinda [51] and Hao et al. [52] have shown that the interaction between economic growth and EQ could be non-linear. Given the expansion of the experimental literature, several studies have been conducted in the EKC framework,

yielding different results. In fact, the results of these studies are even different in similar countries. For instance, although Gozgor and Can [53] supported the EKC in Turkish context, Karahasan and Pinar [54] rejected the framework in the same country. In another research, Shahzad et al. [1] evaluated the association of per capita income and ecological footprint index in the United States during 1965 - 2017, the results validated the EKC. Hao et al. [52], Pata [55], Ozcan et al. [56], and Fakher [14] came to similar findings. Meanwhile, the association of economic development and degradation of the environment followed a U-shaped pattern in a study by Godil et al. [57], inverse N-shaped in a study by Destek and Sinha [58], N-shaped in a study by Shujah-ur Rahman et al. [59], and positive in a study by Khan et al. [6]. In similar study conducted by Sinha and Sengupta [60], the N-shaped EKC model was confirmed for APEC countries. This N-shaped EKC model indicates that after a specific level of income, higher income may once again result in positive associations of economic development and pollution [48]. According to the study of Torras and Boyce [61], because of the decreasing returns on technological changes, the scale impact will overcome the composition and technical impacts. Hence, the N-shaped EKC model happens for economic growth - pollution nexus. This N-shaped relationship is also found for BRICS countries when using GMM approach [35] and for 26 Organization for Economic Co-operation and Development (OECD) countries when using panel data estimation techniques [46].

According to the mentioned EKC literature, the inverted U-shaped is supported by 45% of the studies, whereas the N-shaped and inverted N-shaped EKC were found valid in just 17% and 11% of the cases, correspondingly. This is why the U-shaped relationship accounts for 21% and only 6% of the studies showed linear pattern (positive and negative). These results are illustrated in Figure 3; clearly, it indicates that there are mixed and inconclusive results on economic development – EQ nexus which may be associated with the type of environmental indicator adopted by the researchers.

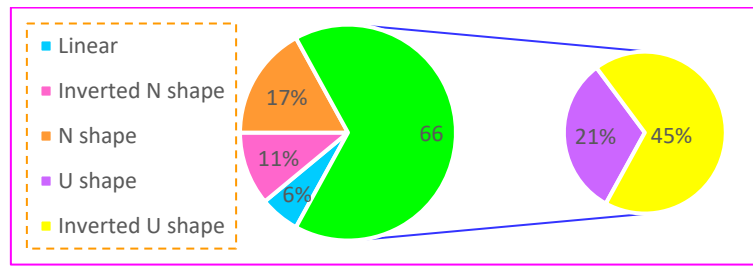


Figure 3. Various relationships between economic development and degradation of environment

Source: Authors' estimations published studies.

The second group involved studies that focused on the association of REC and NEC with environmental deterioration. Relevant literature shows several research carried out on the effects of REC and NEC on degradation of environment. A growing body of this literature have concluded that increased NEC reduces EQ. For example, in a study Ma et al. [4] evaluated the role of NEC in degradation of environment during 1995-2019, and reported that increased degradation of the environment resulted from NEC. In a similar research, Sharif et al. [3] assessed the role of NEC in environmental degradation in Turkey during 1965-2017 with the use of the Quantile Autoregressive Distributed Lag (QARDL). They found that using non-renewable energy led to increased environmental degradation. Similar results were obtained by Saud et al. [7] in the countries of Belt and Road Initiative (BRI) and by Fakher [14] for developing countries. Furthermore, REC has been assessed in many studies owing to its crucial role in EQ. For example, Anwar et al. [47] and Pata [55] evaluated the role of REC on the quality of environment for ASEAN countries and the USA, respectively. According to the results, the use of renewable energies and environmental degradation were negatively correlated. Similarly, Wang and Zhang [62], and Acheampong et al. [63] approved a positive role of REC in the EQ. In another study, Destek and Sinha [58] used second generation panel data methods and confirmed the significant negative linkage of REC and ecologic footprint as a proxy of degradation of the environment in economic Co-operation and development countries during 1980-2014. Similarly, using second generation panel data approaches, Destek et al. [45] supported the previous studies by concluding that REC and EQ were positively and

significantly associated in EU countries; in other words, they obtained a consensus to consume renewable energy to decrease the environmental degradation. On the contrary, in some studies conducted by Menyah and Wolde-Rufael [64] on the US during the period 1960–2007 and Apergis et al. [65] on a group of 19 developed and developing nations from 1984 to 2007, the insignificant associations of REC and environmental degradation is confirmed.

The third group referred to academic researches that the relationship of FD and EFI were appraised in Turkey during 1986-2016, and the results were indicative of the positive impacts of financial situation on the EFI [57]. Additionally, there are other similar studies, which revealed financial situation was positively and significantly related to environmental pollution. For instance, Fakher et al. [1] implemented System Generalized Method of Moment (SYS-GMM) estimator to examine the effects of financial situation on six environmental indicators using a sample drawn from the selected OECD countries during 2010-2019. According to the results, their study concludes that FD affects degradation of environment positively. In line with this result, the correlation of FD and degradation of environment was analyzed in the empirical works of Nathaniel et al. [66] using panel data of the years 1990-2016. The Augmented Mean Group algorithm is used in their study, and based on the results, FD positively contributes to environmental degradation. Inconsistent with the foregoing research, Fakher [14] reported a decrease in the ecological footprint index following the use of the Bayesian Approaches (BA) to evaluate the role FD in environmental degradation while using the ecological footprint index for a period of 1996 to 2016 in developing nations. Similarly, Fakher et al. [18] for OECD countries; Ganda [34] for OECD countries; Ahmed et al. [67] for Malaysian context, and, Acheampong et al. [68] for developed and emerging financial economies, found that FD and environmental degradation were negatively and significantly associated. Meanwhile, the effects of financial situation on environmental degradation were unclear in some researches [e.g., 69 and 70].

The fourth group included pragmatic research evaluating the connection of trade openness (TO) and degradation of environment, which yielded different results. For instance, Destek and Sinha [58] used second-generation panel data methodologies to investigate the impacts of TO on the ecologic footprint index for selected OECD countries in the period of 1980-2014, the results of whom showed the negative effects of TO on degradation of environment. Similar results were obtained by similar research by Sharif et al. [3]. Meanwhile, other research, including the study by Usman et al. [15] and Wang and Zhang [62], reported the increasing effect of TO on environmental degradation.

Finally, the existing literature indicates vague results regarding the influence of POPD on the environment. For instance, Khan et al. [71] reported that population increments emission in the Arab regions. In similar studies, Rahman [72] confirmed that population density significantly intensified emission in India; while in 28 European nations, Pham et al. [73] found that population size accelerates environmental pollution in the long run. Dong et al. [74] and Ali et al. [75] confirmed that population density contributes significantly to emissions. On the other hand, several studies reported contrasting results. For example, Begum et al. [76] discovered that population density has no significant influence on emission in Malaysia. Furthermore, Sulaiman and Abdul-Rahim [77] found that population does not stimulate emissions in Nigeria.

By summarizing the literature related to GDP, REC, NEC, FD, POPD, TO, and EQ, one can see that contradictory results have been obtained by various empirical studies performed in this area. This is currently among the immense challenges in the research on environmental economics, attracting the attention of many researchers in this field. The existing empirical literature on EKC hypothesis have so far focused on one deputy of the quality of environment as the dependent variable in the models and certain academic gaps still exist. In this respect, using a set of important environmental indices simultaneously as dependent variables, which

show the environmental status, is a challenging issue that has been overlooked by previous studies.

Referring back to the abovementioned empirical literature, the present research is distinguished from other studies owing to its innovative objective and implementation method. Firstly, as far as the authors know, most previous studies used an index as a dependent variable in the model to evaluate the N-shaped EKC and the effect of other determining variables on EQ. Meanwhile, the current paper is the first empirical research conducted using six environmental indices consisting of EFI, EPI, ESI, EVI, ANS, and PNI simultaneously, to evaluate the effects of REC, NEC, and GDP on EQ considering the N-shaped EKC in selected countries of OPEC. Secondly, the CTS was applied instead of using the conventional variable of TO (the aggregation of exports and imports over GDP) for the first time. Notably, the CTS was exploited due to a lack of belief in trade openness's ability to fully describe the trade ratio. Thirdly, we evaluated the effect of REC, NEC, FD, POPD, CTS and GDP on EQ and the N-shaped EKC applying DSURE technique.

3. Research data and method

3.1. Data

In this study, annual data covering the period from 1994 to 2019 for 13 selected OPEC countries (including Iran, Iraq, Saudi Arabia, Nigeria, Libya, Angola, Ecuador, Indonesia, Kuwait, Qatar, United Arab Emirates, Venezuela and Algeria) is used. Although OPEC countries overwhelmingly depend on fossil fuels, the consumption of renewables is gradually being embraced in these countries. In Libya, for instance, renewable energy consumption (% of total final energy consumption) increased from 2.48 in 2013 to 2.80 in 2019. In Nigeria, it increased from 82.9 in 2003 to 88.6 in 2009. However, in Algeria, it sailed from 0.07 to 0.19 between 2014 and 2018 [19]. The continuous increase in renewable energy consumption in OPEC countries motivated the need to include REC in equation (1). Besides, the ecological

footprint figures in these countries have further exposed the extent of environmental degradation, as all the selected OPEC countries in this study now occupy an ecological deficit territory [see 78].

Economic growth (GDP per capita) is measured in constant 2010 USD. In addition, the NEC and REC are measured in terms of per capita kg of oil equivalent and % of overall renewable energy in final energy consumption, respectively. Moreover, FD is based on domestic credit to the private sector (% of GDP). Given the general belief that trade openness (TO) alone cannot fully describe trade ratio, following Fakher et al. [2], the combined trade share (CTS) introduced by Squalli and Wilson [79]¹ is used instead of the conventional variable of TO (the aggregated number of imports and exports normalized by GDP). This index considers the share of a nation's trade flow concerning the size of the economy as well as the significance of the trade volume of that country concerning global trade [79]. Also, population density (POPD) are measured in terms of number of people per square kilometre of land area.

EFI was used as an index that expresses the environmental degradation status. Generally, measurement of the EFI takes place through global hectares (Gha) per person. ANS measure involves three types of investment, including physical, human as well as natural capital [26]. PNI comprises four important components of damage by CO₂ and depletion of minerals, energies, and net forest [26]. EPI emphasizes two major goals of protecting the environment and proper natural resources management [27]. ESI evaluates the abilities of different nations concerning environmental preservation for future years. This index is taken from 76 statistical data sets represented through the integration of 21 ESIs. EVI is categorized in five groups, including climatic, geological, geographical, natural resources and services, and human population variables [28]. The World Bank database is accessed to collect data on GDP, FD, and CTS [19], whereas data related to using REC and NEC were derived from the International

¹ See Squalli and Wilson [78], for more details.

Energy Agency (IEA). In addition, various resources were applied to collect data related to the environmental indices, including World Development Indicators [19¹] for ANS and PNI; Yale Center for Environmental Law and Policy² (YCELP) and Columbia University Center for International Earth Science Information Network (CIESIN) for EPI and ESI; South Pacific Applied Geoscience Commission (SOPAC) and the United Nations Environment Programme (UNEP) for EVI; Global Ecological Footprint Network, and National Ecological Footprint Account [80] for EFI.

3.2. Specification of the research model

The N-shaped EKC suggests that the original EKC hypothesis ($\beta_1 > 0$ and $\beta_2 < 0$) will not hold in the long run. Instead, beyond a certain income level, increased income might once again lead to a positive relationship between economic growth and environmental degradation. For this reason, Moomaw and Unruh [81] and Torras and Boyce [61] suggested that the N-shaped relationship occurs when the scale effect overcomes the composition and technical effects. Therefore, they suggested a cubic relationship between economic growth and pollution. For instance, $\beta_1 > 0, \beta_2 < 0$ and $\beta_3 > 0$ it reveals an N-shaped relationship between economic growth and pollution, which is part of our empirical estimation as well.

The present paper seeks to investigate the N-shaped EKC and investigate the role of REC, NEC, FD, and CTS on the quality of the environment. In this regard, following the studies of Fakher and Inglesi-Lotz [29], Ulucak and Yucel [17], and Shahbaz et al. [82], the pragmatic model can be presented in the following form:

$$EI_{it} = f(Y_{it}, Y_{it}^2, Y_{it}^3, NEC_{it}, REC_{it}, FD_{it}, CTS_{it}, POPD_{it}) \quad (1)$$

Where $i = 1 \dots \dots 13$ and $t = 1994 \text{ to } 2019$. Y_{it} , Y_{it}^2 , and Y_{it}^3 indicate per capita gross domestic product, the square of per capita gross domestic product, and cubic form of per capita

¹ See <https://data.worldbank.org>

² See <https://epi.envirocenter.yale.edu/>

gross domestic product, respectively, to evaluate an N-shaped EKC. In addition, NEC_{it} and REC_{it} express non-renewable and renewable energy use, respectively. Moreover, FD_{it} demonstrates the financial development variable, whereas CTS_{it} and El_{it} indicate combined trade share, and environmental indices, respectively. Also, $POPD_{it}$ indicates population density.

Since we will be looking for an N-shaped effect, a cubic regression model is the best fit for this research and can be written as follows:

$$\ln EI_{it} = \beta_0 + \beta_1 \ln Y_{it} + \beta_2 \ln Y_{it}^2 + \beta_3 \ln Y_{it}^3 + \beta_4 \ln NEC_{it} + \beta_5 \ln REC_{it} + \beta_6 \ln FD_{it} + \beta_7 \ln CTS_{it} + \beta_8 POPD_{it} + \omega_{it} \quad (2)$$

The coefficient's signs (+/−) related to Y (square and cubic Y) describe the shape of the curve (for details, see Table 2).

Table 2. The possible scenarios for the shapes of EKC curve (see the study of Shujah-ur-Rahman et al. [59], for more details)

Coefficient's sign (+/−)	Shape of curve
$\beta_{1i} = \beta_{2i} = \beta_{3i} = 0$	No relationship
$\beta_{1i} < 0$ and $\beta_{2i} = \beta_{3i} = 0$	Monotonic decreasing
$\beta_{1i} > 0$ and $\beta_{2i} = \beta_{3i} = 0$	Monotonic increasing
$\beta_{1i} < 0, \beta_{2i} > 0$ and $\beta_{3i} = 0$	U-shaped
$\beta_{1i} > 0, \beta_{2i} < 0$ and $\beta_{3i} = 0$	Inverted U-shaped (EKC hypothesis)
$\beta_{1i} > 0, \beta_{2i} < 0$ and $\beta_{3i} > 0$	N-shaped
$\beta_{1i} < 0, \beta_{2i} > 0$ and $\beta_{3i} < 0$	Inverted N-shaped
$\beta_{1i} > 0, \beta_{2i} > 0$ and $\beta_{3i} < 0$	Cubic polynomial inverted-U shaped
$\beta_{1i} < 0, \beta_{2i} < 0$ and $\beta_{3i} > 0$	Cubic polynomial U-shaped

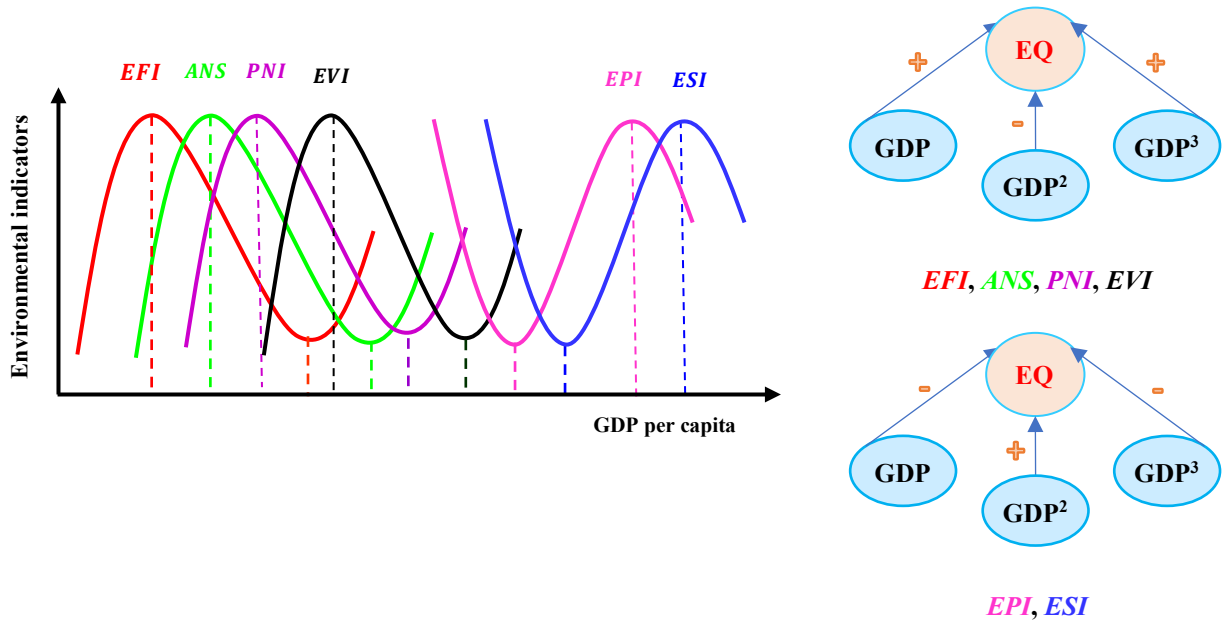


Figure 4. Expected signs between GDP per capita and environmental indicators

In the present research, we applied six environmental indices to evaluate their relationship with independent variables of the model. Given the objective of the research (i.e., evaluation of the N-shaped EKC by considering different environmental indices and comparing the results), six important environmental indices were applied in the model as dependent variables. Accordingly, Eq. (2) is respectively modified to form six models presented from Eqs. (3) to (8) for probing the long-run relations of the studied variables with environmental indices.

$$\ln EFI_{it} = \alpha_0 + \alpha_1 \ln Y_{it} + \alpha_2 \ln Y_{it}^2 + \alpha_3 \ln Y_{it}^3 + \alpha_4 \ln NEC_{it} + \alpha_5 \ln REC_{it} + \alpha_6 \ln FD_{it} + \alpha_7 \ln CTS_{it} + \alpha_8 \ln POPD_{it} + \varepsilon_{it} \quad (3)$$

$$\ln ANS_{it} = \theta_0 + \theta_1 \ln Y_{it} + \theta_2 \ln Y_{it}^2 + \theta_3 \ln Y_{it}^3 + \theta_4 \ln NEC_{it} + \theta_5 \ln REC_{it} + \theta_6 \ln FD_{it} + \theta_7 \ln CTS_{it} + \theta_8 \ln POPD_{it} + \nu_{it} \quad (4)$$

$$\ln PNI_{it} = \lambda_0 + \lambda_1 \ln Y_{it} + \lambda_2 \ln Y_{it}^2 + \lambda_3 \ln Y_{it}^3 + \lambda_4 \ln NEC_{it} + \lambda_5 \ln REC_{it} + \lambda_6 \ln FD_{it} + \lambda_7 \ln CTS_{it} + \lambda_8 \ln POPD_{it} + \vartheta_{it} \quad (5)$$

$$\ln EVI_{it} = \xi_0 + \xi_1 \ln Y_{it} + \xi_2 \ln Y_{it}^2 + \xi_3 \ln Y_{it}^3 + \xi_4 \ln NEC_{it} + \xi_5 \ln REC_{it} + \xi_6 \ln FD_{it} + \xi_7 \ln CTS_{it} + \xi_8 \ln POPD_{it} + \tau_{it} \quad (6)$$

$$\ln EPI_{it} = \varphi_0 + \varphi_1 \ln Y_{it} + \varphi_2 \ln Y_{it}^2 + \varphi_3 \ln Y_{it}^3 + \varphi_4 \ln NEC_{it} + \varphi_5 \ln REC_{it} + \varphi_6 \ln FD_{it} + \varphi_7 \ln CTS_{it} + \varphi_8 \ln POPD_{it} + \pi_{it} \quad (7)$$

$$\ln ESI_{it} = \delta_0 + \delta_1 \ln Y_{it} + \delta_2 \ln Y_{it}^2 + \delta_3 \ln Y_{it}^3 + \delta_4 \ln NEC_{it} + \delta_5 \ln REC_{it} + \delta_6 \ln FD_{it} + \delta_7 \ln CTS_{it} + \delta_8 \ln POPD_{it} + \mu_{it} \quad (8)$$

In each equation above, we evaluated the effect of significant economic variables, including *GDP* per capita, *NEC*, *REC*, *FD*, *POPD* and *CTS* as independent variables on each environmental index including *EFI*, *EPI*, *ESI*, *EVI*, *ANS*, and *PNI* selected as the dependent variable in the models. In order to smoothen the panel data, the analyzed variables were changed to their natural logarithm [83]. In this way, the results provided will have sufficient consistency in comparison with simple linear forms [7]. In the above equations, ecological footprints index ($\ln EFI_{it}$), adjusted net savings ($\ln ANS_{it}$), pressures on nature ($\ln PNI_{it}$), environmental vulnerability ($\ln EVI_{it}$), environmental performances ($\ln EPI_{it}$), environmental sustainability ($\ln ESI_{it}$) are dependent variables. Moreover, GDP per capita ($\ln Y_{it}$), squared GDP per capita ($\ln Y_{it}^2$), cubic GDP per capita ($\ln Y_{it}^3$), non-renewable energy ($\ln NEC_{it}$), renewable energy ($\ln REC_{it}$), financial development ($\ln FD_{it}$), population density ($\ln POPD_{it}$) and composite trade share ($\ln CTS_{it}$) are regressors. i indicates panel countries, t shows the time-span, ε_{it} , ν_{it} , ϑ_{it} , τ_{it} , π_{it} , and μ_{it} are the error term, and α_0 , θ_0 , λ_0 , ξ_0 , φ_0 and δ_0 are the slope intercept in model 1 to model 6, respectively. The research methodology related to the estimation of research models is briefly explained below.

3.3. Estimation strategy

The existence of cross-sectional dependence (CD) in the panel data is among the important issues in panel data econometrics. Accordingly, determining cross-sectional independence is the first step in panel data investigations. In this regard, the Pesaran's [84] CD test was applied in the present research. The dataset of the OPEC nations suffered from CD, and therefore, using the conventional panel unit-root tests might result in unreal unit-root findings. Therefore, the CIPS and CADF were applied to avoid inefficient results caused by CD [67].

In the next stage, it was necessary to scrutinize the presence of long-term associations between the research factors. Hence, the conventional first-generation panel cointegration methods can no longer be used during cross-sectional dependence. Therefore, second-

generation cointegration methods (i.e., Westerlund's [85] cointegration test) is used. To evaluate panel cointegration, Westerlund [85] proposed four statistics of P_t , P_a , G_t , and G_a . The hypothesis of no cointegration was tested against the hypothesis of cointegration by the panel statistics of G_a and G_t . Moreover, the hypothesis of no cointegration was tested concerning the assumption of the presence of at least one cointegration vector by P_a and P_t panel statistics [86]. After this, the long-run coefficients in the models are estimated.

Considering the problems of dependency of cross-sections data, the present paper uses the DSURE approach proposed in Mark et al.'s [87] to study the long-term associations of the variables for the selected OPEC countries. This approach is more proper and robust as compared to the first-generation methods (i.e., fixed or random effect, DOLS, and FMOLS) which led to unpredictable results when cross-sectional dependence issue is present [7].

In case the time (T) exceeds the numbers of cross-sections (N), this estimator (DSURE) is more favorable as compared to other estimators and provides consistent results [88]. The DSURE approach provides robust outcomes amidst CD. Also, it handles the existing concerns pertaining to the data with balanced panel form in which the period of time (T) outpaces the cross-sections' numbers (N) [89]. The conceptual model of the research for assessing the N-shaped EKC while considering six environmental indices is presented in Figure 5.

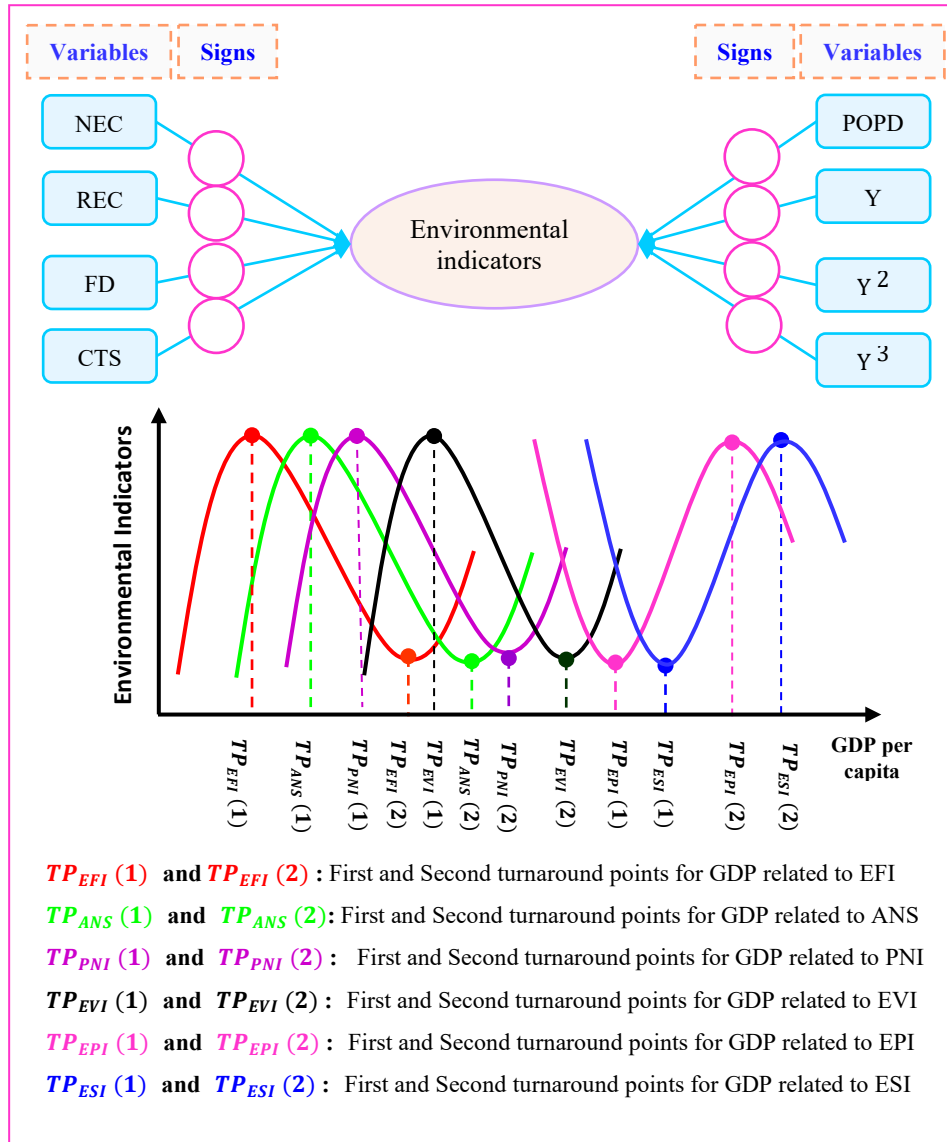


Figure 5. Conceptual model in this research

4. Research empirical results

In this part, the descriptive statistics of variables under study are presented. The findings from the econometric estimation techniques are also presented and discussed in this section.

4.1. Descriptive statistics

Tables 3 and 4 summarize the descriptive statistics of the dependent as well as independent variables for the six (6) countries during the 25-year interval. Based on Table 3, we find that

EFI, PNI and EPI are more volatile compared to other environmental indices. In this group of selected countries, EFI has the most volatility, while the least volatility belongs to EVI.

Table 3. Descriptive statistics results related to environmental indices (dependent variables)

Stat/Var	ANS	EFI	EPI	ESI	EVI	PNI
Mean	2.165215	4.195194	6.819044	36.96436	294.6067	12.78455
Max	5.648086	16.66270	17.71566	71	382	85.55836
Min	0.236617	0.654458	0.000283	14	201	0.913675
Std. Dev.	1.094643	3.930305	5.384240	12.54867	42.38793	10.97835
CV	0.505559	0.936859	0.789589	0.33948	0.14388	0.858720

Note: The units for each of the figures have been mentioned in Section 3.

Moreover, Table 4 reveals that NEC, GDP and REC are most volatile as compared to other variables, respectively. In this group of selected countries, REC has the most volatility, while the least volatility belongs to CTS.

Table 4. Descriptive statistics results related to independent variables

Stat/Var	Y	NEC	REC	FD	CTS	POPD
Mean	16942.54	4058.559	24.02418	26.8837	0.725755	2.926065
Max	69679.09	22120.37	88.83158	135.74	1.759520	17.51095
Min	70.52	433.57	0.06977	0.03276	0.213150	1.022058
Std. Dev.	20485.14	4909.06	31.92114	24.3855	0.296866	2.549223
CV	1.21	1.20	1.33	0.91	0.41	0.87

Note: The units for each of the figures have been mentioned in Section 3.

4.2. Estimation of the empirical models

From the research methods, examining the presence of cross-sectional dependence in the data is the first step in estimating the panel data. As shown in Table 5, it is not possible to accept the null hypothesis concerning lack of cross-sectional dependence. To put it simply, there is some concerns related to dependency of cross-sections in the variables under consideration.

Table 5. Values obtained from cross-section dependence test

Variables	CD	Variables	CD
lnY	3.129 ^a	lnEFI	2.291 ^a
lnNEC	2.465 ^b	EPI	11.235 ^a
lnREC	2.816 ^a	EVI	13.751 ^a
lnFD	9.530 ^a	ESI	20.301 ^a
lnCTS	2.727 ^a	ANS	11.454 ^a
POPD	6.631 ^a	PNI	15.038 ^a

Significance represented by a and b at the levels of 1% and 5% correspondingly.

Due to the presence of cross-sectional dependency among the variables, the application of the second-generation unit root tests such as CIPS unit root technique is necessary to afford decent stationary results. Table 6 displays the findings of CIPS unit root test. The estimates presented in Table 6 imply that the variables under study are integration of order one, I (1).

Table 6. Values obtained by CIPS

Variables	CIPS	
	Levels	First difference
lnY	-2.081	-3.924 ^a
lnNEC	-1.389	-3.517 ^a
lnREC	-1.729	-3.347 ^b
lnFD	-1.834	-3.478 ^a
lnCTS	-2.052	-3.065 ^a
lnEFI	-1.428	-2.874 ^b
lnEPI	-2.233	-3.345 ^a
lnEVI	-2.076	-2.981 ^a
lnESI	-1.212	-3.463 ^a
lnANS	-2.198	-2.702 ^c
lnPNI	-2.086	-3.309 ^a
lnPOPD	-2.136	-3.471 ^a

Significance represented by a, b and c at levels of 1%, 5% and 10%, correspondingly.

After confirming that the variables stationary, the cointegration of the variables needs to be checked. The Westerlund [85] cointegration test is applied since it provides robust and consistent results in the presence of cross-sectional dependence. Table 7 represents the obtained results, showing that it is impossible to accept the null hypothesis of lack of cointegration. In other words, the existence of long-run association of the model variables is accepted and in the final step, the research models can be estimated without worrying about the occurrence of false regression problem.

Table 7. Results of the Westerlund's [85] cointegration test

Statistic	Values	Z-values	P-values
Model 1: $\ln EFI = F(\ln Y, \ln Y^2, \ln Y^3, \ln NEC, \ln REC, \ln FD, \ln CTS, \ln POPD)$			
G_t	-3.430	-2.283	0.011
G_a	-6.418	3.258	0.040
P_t	-8.266	-3.106	0.001
P_a	-6.120	3.204	0.000
Model 2: $\ln ANS = F(\ln Y, \ln Y^2, \ln Y^3, \ln NEC, \ln REC, \ln FD, \ln CTS, \ln POPD)$			
G_t	-5.199	-1.809	0.021
G_a	-5.111	3.108	0.001
P_t	-6.253	-1.568	0.050
P_a	-3.153	2.201	0.009
Model 3: $\ln PNI = F(\ln Y, \ln Y^2, \ln Y^3, \ln NEC, \ln REC, \ln FD, \ln CTS, \ln POPD)$			
G_t	-3.238	-3.566	0.000
G_a	-12.580	-5.303	0.000
P_t	-8.233	-2.671	0.004
P_a	-9.171	-3.306	0.001
Model 4: $\ln EVI = F(\ln Y, \ln Y^2, \ln Y^3, \ln NEC, \ln REC, \ln FD, \ln CTS, \ln POPD)$			

G_t	-10.983	-1.966	0.008
G_a	-18.089	-1.689	0.025
P_t	-2.561	2.903	0.096
P_a	-8.621	2.053	0.040
Model 5: $\ln EPI = F(\ln Y, \ln Y^2, \ln Y^3, \ln NEC, \ln REC, \ln FD, \ln CTS, \ln POPD)$			
G_t	-4.721	-3.799	0.000
G_a	-3.985	2.225	0.023
P_t	-4.023	-3.113	0.001
P_a	-1.776	2.804	0.198
Model 6: $\ln ESI = F(\ln Y, \ln Y^2, \ln Y^3, \ln NEC, \ln REC, \ln FD, \ln CTS, \ln POPD)$			
G_t	-4.077	-4.181	0.000
G_a	-1.288	6.512	1.000
P_t	-5.931	2.512	0.991
P_a	-0.687	4.271	1.000

After establishing the cointegrating among the variables, we estimate the long-run coefficients the DSURE method. Table 8 reports the DSURE estimation results across all the six models (models 1 to 6). Besides, the estimated coefficients and the estimated EKC are separately shown in Figures 7, 8 and 9 for the six environmental indicators. Now, the findings of all these six models are discussed one after the other.

Table 8. Result of panel DSURE estimation

	Model 1		Model 2		Model 3		Model 4		Model 5		Model 6	
Ind. Var.	Dep.	Var. :	Dep.	Var. :	Dep.	Var. :	Dep.	Var. :	Dep.	Var. :	Dep.	Var. :
	EFI		ANS		PNI		EVI		EPI		ESI	
$\ln Y$	1.161 (0.001)		2.028 (0.023)		1.081 (0.000)		1.085 (0.001)		-2.206 (0.003)		-1.231 (0.000)	
$\ln Y^2$	-0.036 (0.000)		-0.022 (0.001)		-0.072 (0.000)		-0.055 (0.002)		1.034 (0.000)		0.098 (0.001)	
$\ln Y^3$	0.019 (0.001)		0.015 (0.000)		0.081 (0.000)		0.071 (0.005)		-0.021 (0.001)		-0.046 (0.000)	
$\ln NEC$	0.124 (0.000)		0.106 (0.000)		0.221 (0.000)		0.307 (0.000)		-0.180 (0.000)		-1.298 (0.001)	

lnREC	-0.109 (0.000)	-0.016 (0.001)	-0.034 (0.000)	-0.017 (0.000)	0.098 (0.003)	0.603 (0.000)
lnFD	0.128 (0.000)	0.212 (0.000)	0.104 (0.000)	0.129 (0.013)	-0.258 (0.000)	-0.205 (0.000)
lnPOPD	0.407 (0.001)	0.260 (0.000)	0.196 (0.003)	0.098 (0.003)	-0.069 (0.002)	-0.106 (0.001)
lnCTS	0.168 (0.001)	0.203 (0.001)	1.109 (0.000)	0.106 (0.000)	-0.241 (0.000)	-1.018 (0.000)
EKC Shape	N-shaped	N-shaped	N-shaped	N-shaped	Inverted N-shaped	Inverted N-shaped
Observations	338	338	338	338	338	338
Diagnostic statistics						
R-square	0.911	0.938	0.820	0.914	0.704	0.803
F-statistics	8.521 (0.000)	65.221 (0.000)	16.551 (0.000)	58.612 (0.000)	9.389 (0.000)	61.105 (0.000)

Notes: Estimated p-values are shown in parentheses.

From Models 1-4 of Table 8, the estimated coefficients show that per capita GDP, per capita GDP squared and per capital GDP cubed respectively have a positive, negative and positive effect on EFI, ANS, PNI and EVI. To put another way, we can conclude that an N-shaped EKC relationship exists between per capita GDP and the environmental variables (EFI, ANS, PNI and EVI). This outcome corroborates the earlier findings of Allard et al. [48] for 74 countries, Barış-Tüzemen et al. [90] for Turkey, and Awan and Azam [91] for G-20 countries. In this form of association, environment pollution will initially increase along with an increasing of GDP to the first threshold point; after that, environment pollution will decline to the second threshold point of GDP. Finally, environment pollution begins escalating again at the third threshold point. This result is in agreement with the empirical works by Shujah-ur-Rahman et al. [59] for sixteen countries, and Sinha and Sengupta [60] for Asia-Pacific Economic Cooperation (APEC) countries; while, these outcomes are contrast with the reports of Shahzad et al. [1] for inverted U-shaped EKC, Godil et al. [57] in the case of U-shaped EKC, Destek and Sinha [58] for inverse N-shaped, Nasreen et al. [50] for linear negative shape and Khan et al. [6] for linear positive shape.

On the other hand, for the 5th model (with EPI as dependent element) and 6th model (with ESI as dependent variable), the coefficient of GDP is negative, the coefficient of squared GDP is positive, and the coefficient of cubic GDP is negative, indicating that an inverted N-shaped EKC is validated. This result is consistent with the outcomes obtained by Anwar et al.'s [47] studies for ASEAN countries and Allard et al. [48] for some countries with different levels of income. While this outcome is not corroborated by Hao et al. [52], Ozcan et al. [56], and Fakher [14] for inverted U-shaped EKC. The results of EKC pattern in the models 1 to 6, represent that the influence of GDP on EQ is not stable and the environmental degradation will eventually occur along with the increasing GDP.

Additionally, the result of REC is significant and positively associated with EQ in all the models (model 1 to model 6). This corroborates the outcomes of Wang and Zhang [62] for 181 countries and Acheampong et al. [63] in the case of sub-Saharan Africa, who approved the positive impact of REC on EQ. Meanwhile, this outcome is not also supported by the studies of Menyah and Wolde-Rufael [64] in the case of the US and Apergis et al. [65] in the case of a group of 19 developed and developing countries, who reported the insignificant linkage between REC and environmental degradation. In contrast, NEC positively influences the environmental degradation based on all the models (model 1 to model 6). The positive environmental effect of NEC on environmental degradation is in agreement with the results reported by Ma et al. [4] in the case of China and Sharif et al. [3] for Turkey context, who claim that an increase in NEC leads to increase in environmental degradation.

For all the models (model 1 to model 6), FD also has a negative and significant impact on EQ, which represents that environmental degradation tends to increase with FD. This finding is congruent with those stated by Fakher et al. [2] in the case of thirty (30) countries and Nathaniel et al. [66] for 13 African nations, Zakaria and Bibi [92] for South Asia, Shahbaz et al. [93] for Pakistan, and in contrast with those mentioned by Acheampong et al. [68], Musa et

al. [31], and Ahmed et al. [67], suggesting the positive effect of FD on the EQ and by Seetanah et al. [70] and Cheng et al.[69], indicating the insignificant effect of FD on EQ.

The estimated coefficient show that CTS has a significant negative effect on EQ in terms of the first 4 models. On the other hand, for the 5th model (with EPI as dependent variable) and 6th model (with ESI as dependent variable), the results show the estimated coefficient of CTS is significantly negative. This outcome agrees with the earlier findings of Usman et al. [15] for the US and Wang and Zhang [62] for 182 countries. Meanwhile, this result contradicts several studies such as Destek and Sinha [58] for OECD countries and Sharif et al. [3] for Turkey. By way of explanation, the unfavorable influence of CTS on the quality of the environment is verified.

Going by the positive sign of POPD and its significance for models 1-4 of Table 8, the deteriorating role of POPD on the quality of the environment is confirmed. Furthermore, for the 5th model (with EPI as dependent variable) and 6th model (with ESI as dependent variable), the unfavorable impact of POPD on the quality of the environment is authenticated. To put it another way, the result of POPD is significant and negatively associated with EQ in all the models (model 1 to model 6). This finding is consistent with Rahman [72], Pham et al. [73], Ali et al. [75] and Dong et al. [74]. Meanwhile, this outcome is not in line with Begum et al. [76] and Sulaiman and Abdul-Rahim [77] in the case of Malaysia and Nigeria respectively. Therefore, the behaviors of all variables in the fifth and sixth models are in line with that of in the case of the above-named four models (models 1 to 4) with except for GDP and its squared and cubed terms. Finally, the summary of estimated results, presented in Table 8, is illustrated in Figures 6 and 7.

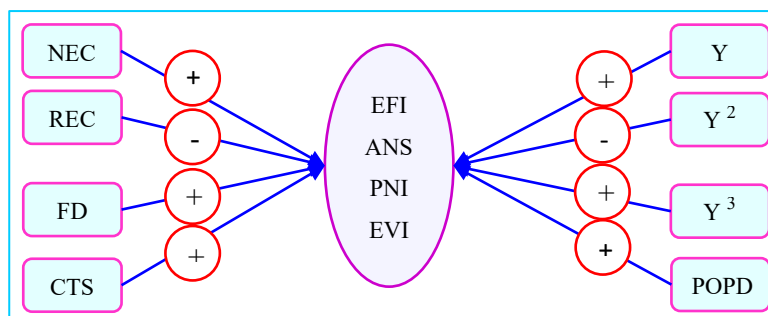


Figure 6. Graphical representation of long-run coefficients estimation related to models 1 to 4

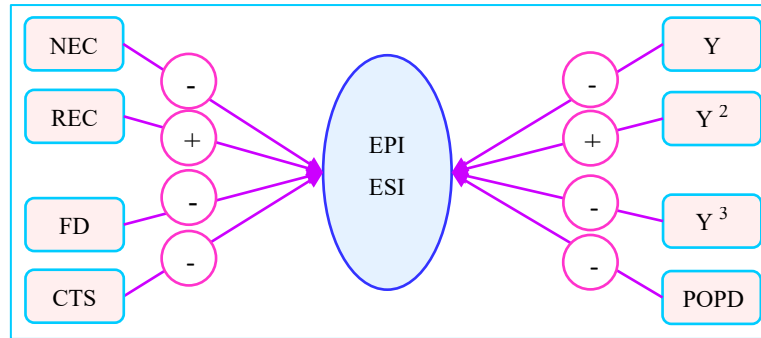


Figure 7. Graphical representation of long-run coefficients estimation related to models 5 and 6

Figure 8 shows a graphical scheme of the results concerning the linkage of environmental indicators, used in this study, with GDP and its squared and cubed terms.

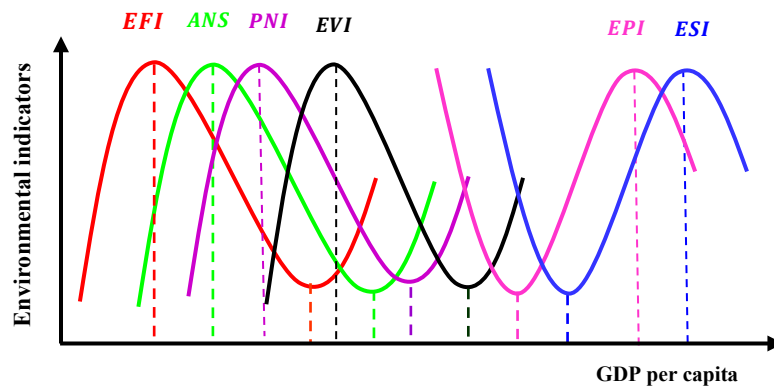


Figure 8 A graphical scheme of the main results

As can be seen from Figure 8, the most significant findings of the analysis show that the N-shaped EKC relationship between per capita income with each of the environmental indicators, such as EFI, ANS, PNI and EVI, is authenticated. On the other hand, with respect to the linkage of environmental indicators, including EPI and ESI with per capita income, an inverted N-shaped EKC is confirmed. As can be seen from Figure 9, the residual plots for 6 estimated models are provided showing that they have the shape of a normal distribution.

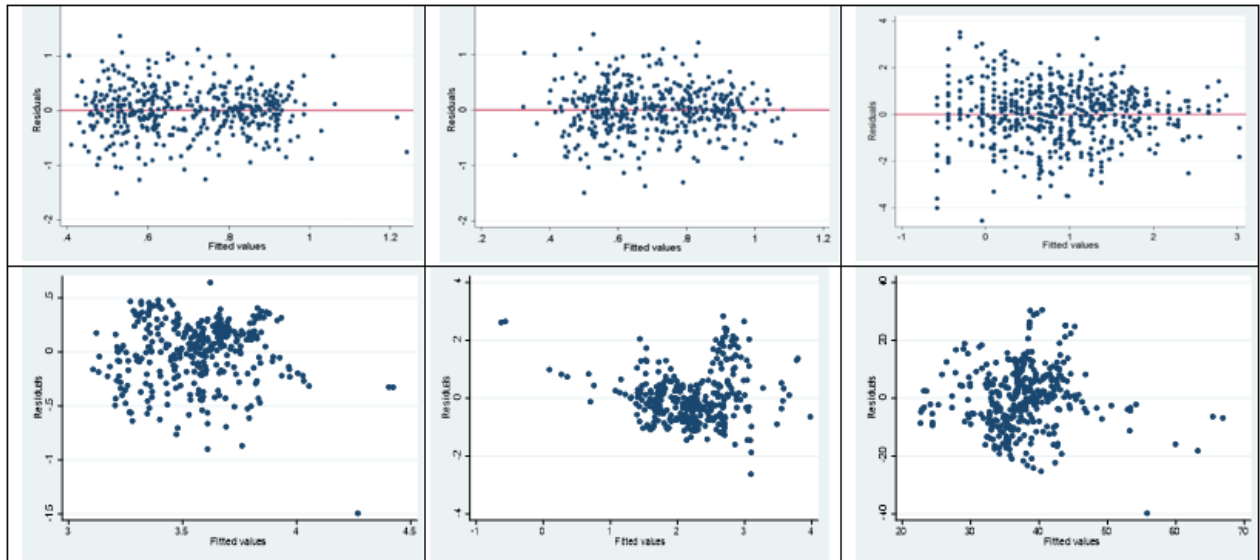


Figure 9 The residual plots for 6 estimated models

5. Conclusion and policy recommendations

This study investigated the effect of non-renewable and renewable energy consumption on selected environmental indicators, within the N-shaped Environmental Kuznets Curve framework, while controlling for composite trade share and financial development. The focus was on OPEC countries, while the time period spans from 1994 to 2019. Besides, we applied various advanced panel econometric techniques, include the Westerlund [85] cointegration test and the Dynamic Seemingly Unrelated Regression Equations (DSURE) that account for core panel data issues ranging from slope heterogeneity to cross-sectional dependence.

Our empirical study represents some important outcomes. First, interestingly, per capita GDP has a N-shaped linkage with each of the environmental indicators, such as EFI, ANS, PN and EVI; Second, an inverse N-shaped EKC in the linkage of environmental indicators including EPI and ESI with per capita income is found; Third, for all the six environmental indicators, NEC reduces the quality of the environment, whereas REC seems to improve the quality of the environment. Fourth. in connection with the all environmental indicators, FD adversely affects EQ. Finally, it was found that with respect to all environmental indicators, CTS and POPD significantly degrade the environment.

The findings from this study have practical implications for environmental sustainability policies in OPEC. The validation of the N-shaped curve indicates that environmental pollution would not decrease by continuous economic growth. However, policymakers are advised to design and implement coherent environmental and economic policies. In other words, we recommend policymakers to design and implement structural policies that would decouple the economy from the environment. We also recommend that OPEC members must implement policies that will hasten the development and adoption of renewable energy. Transition towards renewable energy economy is at the forefront of mitigating environmental pollution and achieving carbon neutrality. Transitioning towards renewable energy economy requires government to develop or reform existing energy policies to prioritize to renewable energy investment and incentives entities that want to venture into the development of renewable energy technologies. Regarding to the expected increase in population density, the policymakers are suggested to encourage the advantages of greater acceptance of the sustainable way of life through water savings and conservation, renewable energy and energy efficiency adoption, green consumables and beverages among others.

6. Limitations and future directions

There are some limitations in this study that could be addressed in future research. Due to space limitations, it was impossible to investigate the causal associations of economic variables and six environmental indicators used in the research models. On the other hand, as data were unavailable in the study period, the present study considered a number of countries (chosen OPEC countries). Accordingly, it is suggested to consider three groups of developed, less developed and underdeveloped contexts and compare the results in future studies. Furthermore, it is suggested to examine the EKC hypothesis considering the six environmental indicators used in this study applying the other appropriate econometric methods such as the Panel Smooth Transition Regression (PSTR) method.

Credit author statement

Hossein Ali Fakher: Conceptualization; Data curation; Empirical analysis; Writing original manuscript; Writing, reviewing and editing; Methodology **Zahoor Ahmed:** Writing original manuscript; writing, reviewing, and editing **Alex O. Acheampong:** Writing and analysis - review; **Solomon Prince Nathaniel:** Writing, reviewing, and editing

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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